



Cardiopulmonary Syndromes (PDQ®)–Health Professional Version

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Overview

Patients with cancer often have comorbid medical problems. Patients older than 65 years bear a disproportionate burden of cancer as well as increased prevalence of medical problems such as chronic obstructive pulmonary disease, heart disease, diabetes, and hypertension.[1] In both primary care and cancer care settings, unexplained symptoms often frustrate physicians and patients.[2] Because many advanced cancers spread to the thorax, symptoms such as dyspnea, cough, chest pain, and palpitations are common. Sorting out the likely cause of these problems and developing appropriate interventions can be difficult. Evidence-based recommendations describing various approaches to cancer-related fatigue, anorexia, depression, and dyspnea have been published.[3] Cancer patients are also often at higher risk of developing pulmonary infections.

Clinicians caring for cancer patients should be familiar with the assessment and treatment of common conditions that manifest as chest symptoms. In addition, these clinicians need to be familiar with some cancer-specific aspects of chest symptoms and syndromes. Dyspnea is a common symptom of certain cancers such as lung cancer and is also common in patients with numerous advanced cancers. Dyspnea is often multifactorial. Optimal treatment requires an understanding of contributing etiologies and pathophysiologies to direct therapeutic interventions as clinically appropriate.

Questions?

Important cardiopulmonary syndromes include the following:

- Malignant pleural effusion.
- Malignant pericardial effusion.
- Superior vena cava syndrome.
- Lymphangitic carcinomatosis.

In this summary, unless otherwise stated, evidence and practice issues as they relate to adults are discussed. The evidence and application to practice related to children may differ significantly from information related to adults. When specific information about the care of children is available, it is summarized under its own heading.

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Dyspnea in Patients With Advanced Cancer

Introduction

Dyspnea is an uncomfortable awareness of breathing. It is a subjective experience involving many factors that affect the quality and intensity of its perception. Patients with comparable degrees of functional lung impairment and disease burden may describe varying intensities of dyspnea, using various words and phrases such as "tightness" and "suffocating."^[1]

Reports on the frequency of dyspnea also vary, depending on the setting and the extent of disease.^[2] In one study, 49% of a general cancer population reported breathlessness, and 20% rated their breathlessness as moderate to severe.^[3] Patients with advanced cancer experience dyspnea more frequently and more intensely than patients with limited disease. One study found that 75 of 135 patients with advanced cancer reporting to an outpatient palliative care clinic had moderate-to-severe dyspnea.^[4] In addition, 60% of 289 patients with lung cancer reported breathlessness at presentation.^[5] Results of a large study showed that 70% of patients suffered from dyspnea in the last 6 weeks of life.^[6] About one-third of patients who could report the intensity of their dyspnea rated it as moderate to severe. Another study revealed that one-half of patients with advanced cancer scored their dyspnea as moderate to severe.^[7]

Pathophysiology and Etiology

The pathophysiological mechanisms of breathlessness are numerous and complex.^[8] Peripheral and central mechanisms, as well as mechanical and chemical pathways, are involved with a variety of sensory afferent sources.^[9-11]

The qualities of dyspnea can be seen as work/effort, tightness, and air hunger. The experience of excess work/effort is caused by sensory-perceptual mechanisms similar to those involved in muscles exercising. Tightness is caused by stimulation of airway receptors with bronchoconstriction. The intensity of air hunger and unsatisfied inspiration is caused by the following:^[12]

- Imbalances of respiratory drive.
- Outgoing signals from the brain.
- Feedback from afferents in the respiratory system.

Dyspnea in patients with advanced cancer has many possible direct causes, and categorizing them can assist in the etiologic work-up. One approach is to divide direct causes into the following groups:

1. **Direct tumor effects** such as intrinsic or extrinsic airway obstruction, pleural involvement, parenchymal involvement by either primary or metastatic disease, superior vena cava syndrome, lymphangitic carcinomatosis, pericardial effusion, and postobstructive pneumonitis.
2. **Indirect tumor effects** such as pneumonia, a pulmonary embolus, paralysis of a hemidiaphragm, or weakening of the respiratory muscles from sarcopenia.
3. **Treatment-related causes** such as pulmonary fibrosis secondary to radiation therapy or chemotherapy, chemotherapy-induced cardiomyopathy, and immunotherapy-related pneumonitis from targeted therapy. Although uncommon, checkpoint inhibitor immunotherapy-related pneumonitis can be potentially severe or fatal.^[13,14] This drug-induced pneumonitis is a diagnosis of exclusion; other differential diagnoses, including infection and worsening malignancy, need to be excluded.
4. **Causes unrelated to cancer** include chronic obstructive airway disease, congestive heart failure, anemia, certain acidotic states, and bronchospasm.
5. **Functional causes** (e.g., anxiety).

One study found that in patients experiencing dyspnea from advanced cancer, a median of five different abnormalities could have contributed to their shortness of breath.^[7] Spirometry was abnormal in 93% of 100 patients examined, with 5% having obstructive patterns, 41% restrictive patterns, and 47% mixed patterns. Also, 49% of patients had lung

cancer, 91% had abnormal chest radiographs, and 65% had parenchymal or pleural involvement. These results show that a subset of patients experience shortness of breath without any apparent lung involvement. The potentially correctable causes of dyspnea included:

- Hypoxia (40%).
- Anemia (20%).
- Bronchospasm (52%).

No significant association between the type of respiratory impairment and the degree of dyspnea was found. Most of these patients were current or former smokers and had a significant lowering of their maximum inspiratory pressures, suggesting severe respiratory muscle dysfunction.[7] This finding was duplicated in another study.[4] Of patients admitted to hospice care, 34% had histories of cardiac disease, and 24% had histories of respiratory disease.[6] Only 39% of terminally ill patients who reported dyspnea had lung or pleural involvement. The etiology of dyspnea could not be clearly identified in approximately one-quarter of patients. Another study found that 49% of lung cancer patients presented with airflow obstruction.[7,15]

Respiratory muscle dysfunction is an underrecognized factor contributing to dyspnea. Causes of respiratory muscle dysfunction include:[16]

- Neuromuscular disease.
- Malnutrition.
- Deficiencies of potassium, magnesium, and inorganic phosphate.

Poor oxygenation, muscle fatigue, abnormal cortisol and catecholamine levels, and circulating cytokines are also implicated.[16]

Although anxiety is commonly thought to be associated with breathlessness, researchers have found that anxiety and shortness of breath do not invariably go together.[7] One study demonstrated that the involvement of the lungs by cancer, anxiety, and poor maximal inspiratory pressures were correlates of the intensity of dyspnea in patients with advanced cancer.[4]

Assessment

The multidimensional nature of dyspnea must be noted in the complicated assessment of this symptom. Patient-reported outcome is the gold standard for assessment of dyspnea. There is no consensus on the best instrument for assessing dyspnea, but some of the tools used include:

- Visual analog and numerical rating scales, which are commonly used.[8,17-19]
- Borg Scale, which is occasionally used.[17]
- Cancer Dyspnea Scale, a multidimensional instrument that has been shown to be valid and reliable in patients with advanced lung cancer. The subscales measure sense of effort (physical dimension), sense of anxiety (psychological dimension), and sense of discomfort.[20]

These tools are limited, however, because they are unidimensional and do not account for the relative contribution of different factors to a patient's perception of breathlessness. Assessment should include the impact of dyspnea on the patient's functional status and recognition of the dynamic component of dyspnea—namely, exertional dyspnea.

Objective signs, such as tachypnea or the use of accessory breathing muscles, frequently do not match a patient's perception of dyspnea and the degree of functional impairment it causes. Numerous factors, including psychosocial issues, may affect a patient's experience of dyspnea. Pulmonary function tests, with few exceptions, play a limited role in the assessment of this syndrome.

A comprehensive history and examination are essential to an accurate assessment of dyspnea.[8,17] The temporal onset, qualities of the symptom, associated symptoms, precipitating and relieving events or activities, and responses to medications should be reviewed. Sudden onset may herald a pulmonary embolism or infection, whereas gradual onset may suggest development of a pleural effusion. A history of obstructive airways or cardiac disease can shed some light on possible underlying causes. Investigations such as measuring oxygen saturation can be useful in determining whether a patient is hypoxic. In the setting of advanced, incurable cancer, arterial blood gases play a limited role.

In the study looking at checkpoint inhibitor immunotherapy-related pneumonitis, one-third of patients were asymptomatic. However, the most common presenting symptoms were dyspnea (53%) and cough (35%).[13]

Melanoma and non-small cell lung cancer were the most common cancers treated in this study. Interestingly, the duration of treatment before the onset of pneumonitis was quite variable, with a median time to onset of 2.8 months (range, 9 days–19 months). In addition, pneumonitis seemed to occur earlier in patients who received combination therapy than in those who received monotherapy (median, 2.7 months vs. 4.6 months).[13]

Diagnostic tests that may help to determine the etiology of dyspnea include the following:[7]

- Chest imaging by radiography.
- Computed tomography.
- Complete blood counts.
- Oxygen saturation at rest and with exercise.
- Pulmonary function tests, to a much lesser extent.

Maximal inspiratory pressure (MIP) measurements may be helpful, particularly if no apparent cause can be found. MIP is a reliable functional test of the strength of the diaphragm and other respiratory muscles. Functional assessments such as the 6-minute walk test and exercise ergometry may also provide valuable information about the severity and impact of dyspnea.[21,22]

Management of Dyspnea

Management of underlying causes

Lack of a clear understanding of the pathophysiological mechanisms underlying dyspnea hampers the clinician's overall ability to effectively manage it.[8,17] As with all symptoms, it is essential to identify and treat the underlying cause(s) of dyspnea, if possible and when appropriate. Underlying causes (some of them potentially reversible) and their treatments include the following:

- **Tumor obstruction** (upper airways, bronchus, postobstructive pneumonitis, or superior vena cava): Radiation therapy, hormone therapy, or chemotherapy for sensitive tumors. Bronchoscope-guided cauterization or laser ablation of intraluminal masses in large airways can be considered when such technology and expertise are available.[23]
- **Carcinomatous lymphangitis**: A trial of corticosteroids, e.g., dexamethasone. The optimal dose is not clear. Chemotherapy should be considered in sensitive cancers.
- **Superior vena cava obstruction**: Chemotherapy for sensitive tumors, radiation therapy, stenting, opioids for dyspnea, and/or steroids as appropriate for the clinical context.
- **Bronchospasms**: Bronchodilators. The indiscriminate use of bronchodilators, particularly in the absence of bronchospasm, is discouraged. The treatment of underlying chronic obstructive disease should be optimized with bronchodilators and inhaled corticosteroids as appropriate, particularly because many patients may be current or former smokers.

- **Postradiation bronchiolitis obliterans:** A rare complication that occurs in 0.8% to 2.9% of women receiving radiation therapy for breast cancer.[24] Patients are usually treated with corticosteroids.
- **Pleural effusions:** Pleural aspiration by thoracentesis or chest drain. Pleurodesis in selected patients with recurrent effusions should be considered. Indwelling catheters are available for patients with recurrent effusions who are not suitable for pleurodesis.
- **Pericardial effusions:** Drainage for immediate relief and possibly intrapericardial chemotherapy or definitive surgery, depending on the clinical context and goals of care.
- **Gross ascites:** Abdominal paracentesis to relieve diaphragmatic splinting.
- **Cardiac failure:** Diuretics and other drugs as appropriate.
- **Chest infections:** Antibiotics and chest physiotherapy when appropriate.
- **Anemia:** Blood transfusions in select patients.
- **Pulmonary embolism:** Anticoagulate when appropriate.
- **Checkpoint inhibitor immunotherapy-related pneumonitis:** For patients with a severity of grade 2 or higher (based on the National Cancer Institute Common Terminology Criteria for Adverse Events [25]), withholding of drug treatment, administration of corticosteroids, and close follow-up. If pneumonitis worsens, additional immunosuppression may be considered, though its benefit is not clear.
- **Anxiety:** Psychotropic medications and psychotherapy.

Symptomatic management

Symptomatic management of dyspnea is based primarily on the following:

- Oxygen therapy.
- Opioids for palliation of dyspnea.
- Treatment of underlying causes (e.g., superimposed infection), when appropriate.

Opioids are an extremely effective treatment for dyspnea in cancer patients, and fear of side effects should not prevent their appropriate use. Most authorities believe that, if used appropriately, opioids do not hasten death in dyspneic cancer patients; rather, they reduce physical and psychological distress and exhaustion, and early use improves quality of life.[17,26,27] Clinically significant hypoventilation after opioid therapy depends largely on the history of previous exposure to opioids and the rate of increase of the opioid dose. As with opioid use in pain management, starting at a regular low dose in opioid-naïve patients, followed by appropriate dose titration, is advised. Opioid therapy for dyspnea is administered similarly to, and often concurrent with, opioid therapy for pain control. Most of the evidence supports the role of opioids in relieving dyspnea in malignant and nonmalignant conditions.[28,29][Level of evidence: I];[30-32][Level of evidence: II]

Anecdotal and experimental evidence suggest a role for nebulized opioid administration in the treatment of dyspnea.[33-35] Opioid receptors are present on sensory nerve endings in the airways; however, nebulization is an inefficient way of administering drugs.[33] Pharmacokinetic studies suggest that the systemic bioavailability of nebulized morphine is extremely poor and erratic, varying from 4% to 8%.[36] Some patients may experience claustrophobia. The evidence does not support the clinical use of nebulized opioids. More clinical trials are needed to better determine the role of this mode of treatment.

Patients who are hypoxic in room air are likely to benefit from oxygen therapy, probably through a decrease in the chemoreceptor input to the respiratory center and the brain cortex. In two controlled trials, cancer patients with dyspnea who were randomly assigned to oxygen therapy in a crossover design showed significant improvement in their dyspnea.[37,38][Level of evidence: I] The role of supplemental oxygen has also been examined in patients without hypoxemia in a large randomized controlled trial. Supplemental oxygen given at 2 L/min did not significantly improve

dyspnea, compared with supplemental air.[39] Therefore, supplemental oxygen is recommended for cancer patients with hypoxemia but not for those without hypoxemia.

Other investigators have examined the effect of other oxygen delivery modalities on dyspnea in cancer patients, such as:[40]

- **High-flow oxygen.** High-flow oxygen devices deliver up to 40 L of humidified and heated oxygen per minute, and may reduce the intensity of dyspnea in patients who do not respond to low-flow oxygen.[40]
- **Noninvasive ventilation with bilevel positive airway pressure (BiPAP).** BiPAP has been shown in several randomized controlled trials to alleviate dyspnea among hospitalized cancer patients,[40,41] particularly in patients with hypoventilation.

These interventions may be reasonable options for patients with hypoxemia and refractory dyspnea despite the use of low-flow supplemental oxygen.

Other options for symptomatic treatment include:

- Methylxanthines.
- Sedatives.
- Tranquilizers.
- Nebulized local anesthetics.
- Antiprostaglandins.

The role of methylxanthines in cancer-related dyspnea is unclear. Chlorpromazine and promethazine have been shown to decrease dyspnea without affecting ventilation in patients without cancer, but their role in cancer-related dyspnea is uncertain. Four out of five randomized controlled trials failed to show any benefit for using benzodiazepines in cancer patients.[17];[42][Level of evidence: I]

One randomized single-blind study suggested that the combination of two scheduled medications (subcutaneous morphine and subcutaneous midazolam) and one as needed (subcutaneous morphine) for episodes of breakthrough dyspnea is more effective than the other evaluated combinations, but this approach requires further study.[43][Level of evidence: I]

The role of benzodiazepines appears to be limited to treatment of dyspnea that is a somatic manifestation of a panic disorder or that is concurrent with severe anxiety. In a randomized placebo-controlled trial of 432 cancer patients with moderate-to-severe dyspnea, 20 mg of buspirone, a nonbenzodiazepine anxiolytic drug, did not improve dyspnea or anxiety, compared with placebo.[44][Level of evidence: I] Buspirone cannot be recommended at this time for the treatment of dyspnea in cancer patients. No evidence supports the use of nebulized local anesthetics for the treatment of breathlessness.

Given the role inflammation may play in cancer and dyspnea, dexamethasone has been used for symptomatic treatment of dyspnea. A randomized controlled trial prospectively evaluated dexamethasone in a highly selected group of ambulatory patients who had an average dyspnea score of 4 or greater (on a numerical scale of 0–10).[45][Level of evidence: I] Patients were randomly assigned to receive dexamethasone 8 mg bid for 1 week, then 4 mg bid for 1 week versus a matching placebo. In-person assessments were taken at baseline, day 7, and day 14, as well as telephone assessments every other day from day 1 to day 28. Assessments included average dyspnea intensity, activity-adjusted dyspnea, and dyspnea unpleasantness via a numerical rating scale. Dyspnea was also measured via maximal inspiratory pressure and spirometry. The trial was terminated at the second preplanned interim analysis for futility. A clinically meaningful improvement in dyspnea was seen in both the placebo and the dexamethasone groups, leading to the conclusion that high-dose dexamethasone should not be given routinely to alleviate cancer-related dyspnea.

Interestingly, the patients who took dexamethasone had significantly better appetite and well-being scores but worse anxiety, depression, and adverse events.

General support measures

In addition to pharmacological therapies, a number of nonpharmacological measures are suggested for the treatment of dyspnea. These measures include:

- Pursed-lip breathing.
- Diaphragmatic breathing and muscle training.
- Cold air directed across the cheek.
- Meditation.
- Relaxation training.
- Biofeedback techniques.
- Psychotherapy.

The effectiveness of these measures in relieving breathlessness appears to be variable.

Current Clinical Trials

Use our [advanced clinical trial search](#) to find NCI-supported cancer clinical trials that are now enrolling patients. The search can be narrowed by location of the trial, type of treatment, name of the drug, and other criteria. [General information](#) about clinical trials is also available.

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Chronic Cough

Chronic coughing may be the source of major suffering for some patients.^[1] Chronic cough can:

- Cause pain.
- Interfere with sleep.
- Aggravate dyspnea.

- Worsen fatigue.

The causes of cough can be classified much like the causes of dyspnea.

For chronic cough in patients receiving palliative care, consider the following differential diagnoses:

- Malignancy-related causes of cough, e.g., direct irritation by the presence of a tumor or tracheo-esophageal fistula.
- Drug-related cough: rule out cough induced by medications such as angiotensin-converting enzyme inhibitors.
- Other noncancer causes of cough: rule out asthma, chronic obstructive pulmonary disease, postnasal drip, reflux issues, and chronic heart failure.

The optimal therapy for chronic cough is treatment of the underlying disorder, such as:

- Radiation therapy for the underlying endobronchial tumor mass irritating the airway.
- Stenting of tracheo-esophageal fistula.
- Drainage of a pleural effusion.
- Corticosteroids for lymphangitic carcinomatosis.

Cough-suppressing agents such as opioids are commonly used. Anecdotal evidence suggests a role for inhaled local anesthetics, which should be used judiciously and sparingly because of their unpleasant taste, suppression of the gag reflex, and anaphylactic reactions to preservatives in these solutions. Opioid and nonopioid antitussives, such as dextromethorphan, may act synergistically, but further studies are required to confirm this hypothesis.[1] Gabapentin was found to be efficacious compared with placebo for chronic refractory cough, although this study did not include cancer patients.[2]

In cases of increased sputum production, expectorants and mucolytics have been used, but the effects have not been well evaluated. Inhaled sodium cromoglycate has shown promise as a safe method of controlling chronic coughing related to lung cancer.[3]

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Malignant Pleural Effusion

Significance

Malignant pleural effusions are a common complication of malignancy, and malignancy is a common cause of pleural effusions in general. Malignancy causes roughly 40% of symptomatic pleural effusions, with congestive heart failure and infection being the other leading causes.[1] The cancers that account for approximately 75% of all malignancy-associated effusions include the following:

- Lung cancer.
- Breast cancer.

- Lymphoma.
- Leukemia.

Significant use of health care resources is attributable to malignant effusions, with approximately 100,000 cases per year diagnosed in the United States and 43 cases detected per 100,000 hospital admissions.[2]

Pathogenesis

The normal pleural fluid space is occupied with approximately 10 mL of fluid with 2 g/dL protein. A pleural effusion is an accumulation of an abnormal amount of fluid between the visceral and parietal pleura of the chest. Normally, pleural fluid is absorbed by pulmonary venous capillaries (80%–90%), with some of it also absorbed by pleural lymphatics. Malignant effusions are usually exudative rather than transudative. Exudative effusions exhibit any one of the following characteristics:[3]

- Pleural fluid to serum protein ratio greater than 0.5.
- Pleural fluid to serum lactate dehydrogenase (LDH) ratio greater than 0.6.
- Pleural fluid LDH greater than two-thirds of the upper limit of serum LDH.

These exudative malignant effusions are generally caused by:

- Pleural metastases.
- Disruption of pulmonary capillary endothelium.
- Malignant obstruction of pleural lymphatics.

Paramalignant effusions may result from chemotherapy, radiation therapy, atelectasis, or lymph node involvement.

Assessment

Common symptoms associated with malignant pleural effusions include:

- Dyspnea.
- Cough.
- Chest discomfort.

About 20% of patients may experience weight loss and malaise. A chest x-ray is most commonly used for radiographic assessment. About 175 mL of pleural fluid will cause a blunted costophrenic angle discernible on chest radiography. A chest computed tomography scan is more sensitive than a simple chest x-ray and is often used for assessment of loculated effusions because, in some instances, up to 500 mL of loculated fluid can be obscured behind the dome of the diaphragm.[1]

Not all pleural effusions detected in cancer patients are malignant effusions. Cancer patients are prone to developing conditions such as:[1]

- Congestive heart failure.
- Pneumonia.
- Pulmonary embolism.
- Malnutrition.
- Associated low serum albumin.

Each of these conditions may cause a symptomatic effusion for which clinical management would substantially differ from management of a malignant effusion. For this reason, cytological assessment is important. Pleural fluid cytology requires a minimum sample of 250 mL. The morphology of cells obtained from the pleural space can be difficult to assess because of mesothelial and macrophage abnormalities. The diagnostic sensitivity of pleural fluid cytology is approximately 65%, with a specificity of 97%.^[1]

- **Flow cytometry** can be applied to these specimens and is often useful, especially for assessment of lymphomas.
- **Thoracoscopy** and **pleural biopsy** are rarely needed for a definitive diagnosis, but these techniques may be useful when routine pleural fluid collection and assessment are difficult because of loculation of the effusion.
- **Thoracoscopy-guided biopsy** is generally performed under local anesthesia and has a yield of more than 80%, with a lower risk of complications than thoracotomy.

Management of Malignant Pleural Effusions

To treat or not to treat

Pleural effusions are generally markers of advanced, unresectable disease or disease progression. The median survival for patients with malignant pleural effusions is around 3 to 4 months.^[4,5] Because a paramalignant effusion resulting from pneumonia or atelectasis may be present, the cytology should be confirmed before making major treatment decisions. The management strategy depends on the underlying primary malignancy and the number and type of previous therapies. For example, patients with newly diagnosed small cell carcinoma or malignant lymphoma are likely to respond to systemic chemotherapy; however, patients with gastric or ovarian cancer in whom several lines of chemotherapy have failed are unlikely to obtain significant palliation with systemic therapy.

About three-quarters of patients exhibit symptoms such as cough, dyspnea, and chest discomfort. Such patients may benefit from efforts to reduce the fluid burden, depending on the following:

- Their performance status.
- Their expected survival.
- Their preference for risks versus benefits.

The literature on the efficacy of treatment for pleural effusions is difficult to interpret because of the paucity of randomized trials, and wide variability in the response criteria and the timing and duration of follow-up in uncontrolled trials.^[6,7] Generally, the goal of therapy is palliation of symptoms. Measures of success may include:

- Complete drainage of the effusion.
- Lung re-expansion.
- Lack of fluid reaccumulation (i.e., duration of response).
- Subjective report of symptom relief.

The choice of treatment depends on patient prognosis, functional status, and goals of care.

Thoracentesis

Thoracentesis involves percutaneous insertion of a needle for drainage of the effusion. Thoracentesis is not expected to permanently resolve the problem but rather to alleviate acute and severe symptoms. The use of thoracentesis is also appropriate as a therapeutic trial to determine whether fluid drainage is beneficial when the relationship between symptoms and effusion is unclear.

Most effusions will reaccumulate a few days after thoracentesis. The reaccumulation rate is approximately 98% by day 30.^[8] Repeated thoracenteses carry the potential risks of bleeding, infection, and pneumothorax. Other potential

complications of thoracentesis include noncardiogenic pulmonary edema from rapid lung re-expansion (usually with the rapid removal of >1,500 mL) and pleural shock caused by an excessive vagal response to penetration of the parietal pleura. Any of these complications may be lethal, especially for the cancer patient with poor cardiopulmonary reserve.

Chronic long-term indwelling tunneled pleural catheters

Indwelling pleural catheters (IPCs) represent an alternative to pleurodesis for patients with malignant pleural effusion whose dyspnea has responded to thoracentesis. IPCs are relatively contraindicated in patients with a short life expectancy, pleural infections, multiloculated collections, and chylothorax. The insertion of chronic long-term indwelling tunneled pleural catheters is useful to treat recurrent and symptomatic malignant pleural effusions, including for patients with trapped lung. These tunneled pleural catheters allow up to 96% of patients to achieve symptom improvement, with spontaneous pleurodesis occurring in up to 44% of patients.[9] Published results indicate significantly shorter hospital stays for patients with IPCs versus those receiving doxycycline pleurodesis (1 day vs. 6 days). In the IPC group, spontaneous pleurodesis was achieved in 42 of 91 patients. Both the IPC group and doxycycline pleurodesis group reported modest improvement in quality of life (QOL) and dyspnea.[10]

A randomized controlled trial comparing IPC and talc pleurodesis showed similar reduction of dyspnea (24 mm of 100 mm) and similar QOL.[11] IPC use was associated with a shorter initial hospitalization and lower rates of re-treatment, with a spontaneous pleurodesis rate of 51%. However, IPCs also had higher rates of adverse effects, such as infections and catheter blockage. A second randomized controlled trial compared the effect of IPC versus talc pleurodesis on the total number of hospitalization days from procedure to death or to 12 months.[12] There was a statistically significant reduction in the number of hospitalization days for patients with IPCs compared with those who received talc pleurodesis (median, 10 days [interquartile range (IQR), 3–17] vs. 12 days [IQR, 7–21], respectively; $P = .03$). IPC use also resulted in fewer hospitalization days related to pleural effusion management and fewer additional invasive pleural drainage procedures. There were no significant differences in improvement in breathlessness or QOL between the two procedures. The choice between IPC and pleurodesis should be based on patient preference and local resource availability.

Use of pleural sclerosing agents after chest tube drainage

Chemical sclerosants may be administered through a chest tube to create inflammation and subsequent fusion of the parietal and visceral pleura so that fluid cannot reaccumulate in this potential space. Numerous chemical agents can cause the irritation necessary to produce this kind of fusion, called pleurodesis. The ideal agent would produce effective pleurodesis with minimal cost and minimal side effects. Agents that have been studied include:

- Chemotherapeutic agents (bleomycin, cisplatin, etoposide, doxorubicin, mitomycin-C, fluorouracil).
- Antibiotics (doxycycline, minocycline, tetracycline).
- Infectious agents (*Corynebacterium parvum*).
- Biological agents (interferon beta, interleukin-2).
- Bovine dermal collagen.[13][[Level of evidence: II](#)]
- Other agents (talc, methylprednisolone).

Several uncontrolled trials and case series have reported the efficacy of pleurodesis,[14];[15,16][[Level of evidence: II](#)]; [17,18] as have numerous randomized trials.[19-25][[Level of evidence: I](#)] A meta-analysis of pleurodesis studies that were reported between 1966 and 1992 indicated that about two-thirds of patients responded to pleurodesis, and that the most effective agents appeared to be tetracyclines (or tetracycline replacement agents, such as doxycycline and minocycline), bleomycin, and talc.[26]

A prospective, randomized study of video-assisted thoracoscopic pleurodesis with talc versus doxycycline in 33 patients with malignant pleural effusion suggested that talc provides superior short-term and long-term results.[27][[Level of evidence: I](#)] Talc appears to be the least expensive agent, at least when given as a slurry rather than by video-assisted

thoroscopic talc insufflation.[25][[Level of evidence: I](#)] However, bleomycin is the only agent approved by the U.S. Food and Drug Administration for the prevention of recurrent pleural effusions.[1] An observational cohort study investigated the use of intrapleural urokinase in 48 patients with loculated pleural effusions or trapped lungs. Lung reexpansion and resolution of dyspnea occurred in approximately 60% of patients, suggesting that intrapleural urokinase may be useful in treating loculated pleural effusions or trapped lungs in medically inoperable patients with cancer. Most responders maintained pleurodesis when urokinase was followed by minocycline pleurodesis.[28][[Level of evidence: II](#)]

Surgical treatment

For rare patients, standard management of the malignant effusion is unsuccessful and aggressive treatment remains appropriate. Pleuroperitoneal shunting can be considered for these patients. This procedure involves implantation of a shunt with one-way valves that allow the transfer of fluid from the pleural space to the peritoneal space, in which the fluid creates less hazard and is more easily removed. Another option is surgical pleurectomy, which requires general anesthesia. The risk of significant acute and chronic pain as well as other morbidity approaches 20% to 25%, and the risk of mortality at 1 month is 5% to 10%.[2]

Current Clinical Trials

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Malignant Pericardial Effusion

Malignant pericardial effusions occur in up to 21% of cancer patients [1-3] and are frequently not suspected until clinical signs or symptoms of pericardial tamponade develop.[4] Two-thirds of patients have subclinical pericardial effusions with no overt cardiovascular signs or symptoms.[5,6] One-half of cases of pericardial effusion initially present with symptoms of cardiac tamponade.[7] In 50% of cases, pericardial effusion is the first sign of malignant disease.[8] Symptoms of pericardial effusion are often attributed to the underlying cancer and may include the following:[9]

- Dyspnea.
- Fatigue.
- Asthenia.

Symptomatic pericardial effusions are often a preterminal event; however, significant symptom palliation can be achieved with prompt diagnosis and management.

Of patients with malignant pericardial effusions, 50% will have concomitant pleural effusions, and one-third will have pulmonary parenchymal disease.[4]

One-third of patients with pericardial metastases will eventually die of pericardial tamponade.[4] Pericardial involvement contributed to the cause of death in 85% of patients in a series reported in 1962 but in only 46% of patients in a more recent study.[10] Improvements in diagnostic and therapeutic options account for the decrease in mortality over the past 40 years.

Incidence and Prevalence

Malignant pericardial effusion occurs in up to 21% of autopsy cases in patients with common malignancies.[4,7] Of patients with lung cancer, 33% have pericardial metastases at autopsy, and one-third of cases of pericardial metastases are caused by lung cancer. Breast cancer causes 25% of pericardial effusions, and about 25% of patients with breast cancer have pericardial effusions. Hematological malignancies (leukemia, Hodgkin disease, non-Hodgkin lymphoma) cause 15% of cases of malignant pericardial effusions.[11]

A retrospective review of 23,592 effusions over a 24-year period revealed 65 malignant effusions (17%) out of 375 pericardial effusions. Lung cancer was the most common cancer found among the malignant pericardial effusions in males, and breast cancer was the most common in females. In 43% of cases, pericardial effusion was the first detected sign of cancer. Of patients diagnosed with malignant pericardial effusions, 86% died within 1 year of diagnosis, with nearly one-third dying within the first month.[8]

In a study of 31 patients with both cancer and pericardial effusions, malignant pericardial effusion accounted for 58% of the effusions, benign idiopathic pericarditis caused 32% of the cases, and radiation pericarditis caused 10% of cases. [11,12]

Pathophysiology

Malignant involvement of the pericardium is the most common reason for development of pericardial effusions, which result from blockage of venous and lymphatic circulation of pericardial fluid. Such blockage may be caused by primary malignancy of the pericardium, such as pericardial mesothelioma, or by tumors arising in the myocardium, including angiosarcoma, rhabdomyosarcoma, and malignant fibrous histiocytosis. Malignancies can also involve the pericardium through direct extension from carcinomas of the lung or esophagus, thymoma, or lymphoma.[9] Lymphatic or hematogenous metastasis to the pericardium occurs most commonly with the following cancers:

- Carcinomas of the breast and lung.
- Leukemia.
- Lymphoma.
- Melanoma.

Primary tumors of the pleura or pericardium have been termed primary intrathoracic malignant effusions.[13]

Nonmalignant causes of pericardial effusion include:[14-16]

- Pericarditis.

- Myocardial infarction.
- Uremia.
- Hypothyroidism.
- Systemic lupus erythematosus.
- Trauma.
- Postsurgical pericardiotomy syndrome.
- Intrapericardial hematomas.

AIDS may also cause pericardial effusion with pericarditis.[17] Radiation therapy or chemotherapy drugs can cause pericarditis without metastatic involvement of the pericardium. Radiation pericarditis is usually associated with radiation doses to the cardiac window exceeding 30 Gy [10] and occurs most frequently in patients who have received mediastinal radiation for Hodgkin lymphoma or breast cancer.[10] Doxorubicin and cyclophosphamide have been associated with the development of acute pericarditis with effusions.[11,12] Other drugs that may cause acute pericarditis include procainamide, hydralazine, isoniazid, methysergide, phenytoin, and anticoagulants.

Pericardial tamponade results from progressive fluid accumulation in the pericardial sac, causing the following:[15]

- Elevated intrapericardial pressure.
- Diminished stroke volume.
- Decreased cardiac output.
- Progressive decrease in cardiac diastolic filling.
- Hemodynamic compromise resulting in death if not treated.

Hemodynamic compromise occurs when the normal amount of pericardial fluid (approximately 15–50 mL) increases to 200 mL to 1,800 mL.[15,18] When fluid accumulates rapidly, as little as 250 mL of fluid can result in tamponade.[11,19]

Dyspnea occurs in 93% of patients with pericardial effusions.[6] Common symptoms include the following:

- Cough.
- Chest pain.
- Orthopnea (discomfort with breathing while lying flat).

Other symptoms of pericardial effusion include:

- Upper abdominal distention or pressure caused by downward hepatic distention.
- Hiccups resulting from pressure on the diaphragm.
- Pleuritic pain caused by stretching of the pericardium (especially when lying flat).

Signs of effusion include the following:

- Kussmaul's sign (increased distention of jugular veins with inspiration).
- Friedreich's sign (rapid diastolic descent of the venous pulse).
- Pulsus paradoxus (decrease of more than 10 mm Hg in the diastolic pressure on inspiration).

Pericardial friction rubs and fever are more commonly associated with nonmalignant causes of pericardial effusions than with malignant etiologies.[9]

Signs of pericardial tamponade include:

- Tachycardia.
- Pulsus paradoxus.
- Elevated jugular venous pressure.
- Hypotension.

However, some patients may develop tamponade without this clinical pattern.[4]

Diagnosis

A chest x-ray may show widening of the cardiac silhouette [7] if the amount of pericardial fluid collection exceeds 250 mL.[20] However, a chest x-ray cannot determine the degree of cardiac dysfunction or tamponade. Loculated pericardial effusions may not be apparent on standard posterior/anterior or lateral chest radiographic views.[15]

Transthoracic echocardiography using apical, subxiphoid, and parasternal views can evaluate the presence, quantity, and quality of suspected pericardial effusions, as well as associated pericardial masses and inflammation. Moderate effusions on echocardiography show an echo-free space of 10 mm to 20 mm during diastole in M-mode or 2-dimensional echocardiography, whereas severe effusions have an echo-free space exceeding 20 mm.[21,22] Echocardiography can also determine right and left ventricular function and the possibility of right ventricular or atrial diastolic collapse.[7] Left ventricular collapse caused by large pleural effusions without clinically significant pericardial effusions has been reported.[4,16,23,24] However, transesophageal echocardiography may be useful for loculated effusions resulting from adhesions adjacent to the atria, where the thinness of the atrial wall may not be well visualized on transthoracic echocardiography.[4,16]

Echocardiography in pericardial effusion with tamponade shows right atrial or right ventricular compression, or left atrial compression, decreased left ventricular dimension, and absence of collapse of the inferior vena cava on deep inspiration.[6,25] Echocardiography findings predictive of pericardial tamponade have been reported.[26] Right atrial collapse has a sensitivity of 55% to 60% and a specificity of 50% to 68%. Right ventricular diastolic collapse has a lower sensitivity of 38% to 48% but a higher specificity ranging from 84% to 100%. Because neither finding provides 100% sensitivity and specificity, patients who are clinically symptomatic should have a diagnostic pericardiocentesis, even in the absence of definitive findings on echocardiography.[4,27] One study found right atrial collapse present in only 42% of patients and right ventricular collapse in 62%.[27] Nonetheless, 80% of patients with malignant pericardial effusions had symptomatic relief following pericardiocentesis.

The most definitive test for the diagnosis of cardiac tamponade is equalization of diastolic pressures between all cardiac chambers on right-heart cardiac catheterization.[7] However, this invasive technique is not necessary to diagnose tamponade.

An electrocardiogram (ECG) in patients with pericardial effusions typically shows diminished QRS amplitude in all leads. A classic but uncommonly seen finding in large effusions with pericardial tamponade is variation in the amplitude of the P wave and QRS complex in successive beats on ECG, referred to as electrical alternans. This finding results from movement of the heart within the pericardial sac.[6] Electrocardiography is not sufficiently sensitive to diagnose pericardial effusions.

Pericardial fluid cytology has an accuracy of 80% to 90% in diagnosing malignant pericardial effusion.[6,28] Lymphomas and mesothelioma have higher false-negative detection rates on cytology evaluation.[6,29] Pericardial fluid cytology has a specificity of up to 100%, but sensitivity ranges from 57% to 100% [10][[Level of evidence: II](#)] in patients with a known cancer diagnosis and pericardial fluid. Because nonmalignant causes of pericardial effusion can occur in 42% to 62% of patients with cancer and pericardial fluid, a negative cytology examination of pericardial fluid does not help distinguish malignant from nonmalignant causes. The use of more than one cytological preparation (such as concentrating the sample via cytospin, using special markers, or analyzing DNA content) increases the yield over a single preparation.

However, multiple samples using the same technique did not significantly increase the diagnostic yield in a retrospective study of 215 patients.[30] In a survey of 80 samples, measurement of DNA index via flow cytometry of pericardial fluid has a sensitivity of 94.8% and a specificity of 100%, compared with routine cytology, which has a sensitivity of 98.5% and a specificity of 92.3%.[31][[Level of evidence: II](#)]

Pericardial biopsy may increase the sensitivity of diagnosing pericardial effusions of malignant origin. However, because pericardial effusions usually occur in advanced disease and portend a shorter survival than other sites of metastatic involvement, the relief of symptoms rather than diagnosis should be the overriding factor in determining the extent of the evaluation and the course of treatment. Two studies failed to show a difference in survival in cancer patients with pericardial effusion dependent on the results of fluid cytology.[10,32]

In a study of patients with stage I esophageal cancer who underwent radiation and chemotherapy, risk factors for developing pericardial effusion included advanced age, higher pericardial volume 30 (≥ 41.6 percentage of cardiac volume receiving more than 30 Gy), high body mass index, and diabetes mellitus.[33]

Treatment

No large controlled, randomized, prospective clinical trials demonstrate the optimal management of malignant pericardial effusions or tamponade. Treatment should therefore be individualized to maximize symptom relief with minimal impact on quality of life. Treatment options include the following:[34][[Level of evidence: III](#)]

- Percutaneous pericardiocentesis.
- Percutaneous balloon pericardiotomy.
- Pericardial sclerosis.
- Subxiphoid pericardial window.
- Pericardiectomy.
- Pericardiotomy by thoracotomy or video-assisted thoracoscopy.

In choosing a treatment, consider the following:[34][[Level of evidence: III](#)]

- Relief of tamponade.
- Minimal invasiveness.
- Cost.
- Morbidity.
- Safety.
- Shortened hospitalization for patients with advanced disease.
- The patient's prognosis.

Large, symptomatic, malignant pericardial effusions are managed by draining the fluid, unless the goals of therapy dictate a less invasive, conservative approach, with concomitant shorter survival balanced against quality-of-life concerns. If treatment is indicated for tamponade, percutaneous subxiphoid pericardiocentesis is the treatment of choice in the acute setting. Echocardiography is recommended for catheter guidance.[6,35] Catheter drainage is recommended for large effusions to prevent rapid reaccumulation of fluid and subsequent tamponade and for the anticipated survival of the patient.

Recurrent pericardial effusion occurs in 21% [36] to 50% [34,35] of patients following pericardiocentesis. Limited case series suggest rates of pericardial fluid reaccumulation at 30 days ranging from 5% to 33% after pericardial drainage followed by intrapericardial treatment with sclerosing agents or phosphorus-colloid, compared with more than 50% for patients treated with pericardial drainage alone.[34,35]

Prolonged catheter drainage can be effective in preventing fluid reaccumulation; however, the mechanism is unclear. One series reported recurrence in 30% of patients at a median time of 39 days. In another series, the reported recurrence rate of the pericardial effusion was 13% by 1 year of follow-up.[37,38]

The prolonged catheter drainage could be left in for several days.[38,39] The catheter should be left *in situ* until the drainage is minimal (<25–50 mL in a 24-hour period) to none. In one series of 171 patients with malignant pericardial effusion who received echocardiography-guided pericardiocentesis followed by extended catheter drainage, the average time to very little catheter output (<50 mL in 24 hours) was about 3 days.[39] Other treatment options to prevent reaccumulation include intrapericardial sclerosis to obliterate the space within the pericardial sac or pericardiotomy to increase the quantity of fluid drained from the pericardium.

The most effective sclerosing agent for malignant pericardial effusions had been tetracycline, with success rates of up to 80%.[6] However, this agent is no longer available as an intravenous drug in the United States. Alternative sclerosants that have been used include the following:

- Doxycycline.[40]
- Bleomycin, thiotepa.[35];[41][Level of evidence: II]
- Carboplatin.[42][Level of evidence: II]
- Mitoxantrone.[43]
- Docetaxel.
- Chromic phosphate P 32.

Most patients may require three or more treatments to achieve adequate sclerosis.[6] Significant pain is reported by 16% of patients undergoing pericardial sclerosis.[6] The side effects of various sclerosing agents (e.g., chest pain and arrhythmias) must be considered. Of patients undergoing pericardial sclerotherapy, 70% to 80% have no fluid reaccumulation within 30 days of the procedure.[35]

A retrospective comparison of pericardiocentesis with sclerotherapy to open surgical drainage in 60 patients showed similar rates of treatment complications, incidence of recurrent effusion, and survival following treatment in both groups.[44] A retrospective review of 59 patients found similar success rates, whether patients were managed with surgical subxiphoid pericardial window or by pericardiocentesis with or without sclerosis.[34] Patients who underwent pericardiocentesis followed by pericardial window had the longest survival, with a median of 6 months. However, selection bias toward patients with better performance status undergoing more aggressive surgical techniques may contribute this reported survival advantage. The surgical procedure group had significantly higher average costs of \$4,830 compared with \$1,625 for patients managed with pericardiocentesis.[34]

Other studies have reported mortality, recurrence, and survival rates for sclerosis that are similar to or slightly lower than those for subxiphoid window or video-assisted thoracoscopy.[44,45];[46][Level of evidence: II][34][Level of evidence: III] Pericardiocentesis with or without sclerotherapy should be considered instead of more invasive procedures in patients with advanced disease or poor functional status.[47]

Transcutaneous balloon pericardiostomy is another technique that is less invasive than open surgical approaches, which include subxiphoid pericardial windows, thoracotomy with pericardiopleural window formation,[48] and thoracotomy with pericardectomy.

Video pericardioscopy has a diagnostic sensitivity of 97% for detecting malignant effusions.[49] Pericardioscopy also is useful for drainage of loculated effusions.[50][Level of evidence: II] Video-assisted thoracoscopy is preferable to more invasive surgical management and should be considered for patients requiring repeated pericardiocentesis to control symptomatic effusions.[47]

Current Clinical Trials

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Superior Vena Cava Syndrome

Overview

Superior vena cava syndrome (SVCS) is an array of symptoms caused by the impairment of blood flow through the superior vena cava (SVC) to the right atrium. Symptoms that suggest this syndrome include the following:[1]

- Dyspnea.
- Coughing.
- Swelling of the face, neck, upper trunk, and extremities.

In rare instances, patients may complain of hoarseness, chest pain, dysphagia, and hemoptysis.

Physical signs that may be noted on presentation include the following:

- Neck vein distention.
- Thoracic vein distention.
- Edema of the face or upper extremities.
- Plethora.
- Tachypnea.

Rarely, cyanosis, Horner syndrome, and a paralyzed vocal cord may also be present.[1]

SVCS is usually a sign of locally advanced bronchogenic carcinoma. Survival depends on the status of the patient's disease. When small cell bronchogenic carcinoma is treated with chemotherapy, the median survival times with or without SVCS are almost identical (42 weeks or 40 weeks, respectively). The 24-month survival rate is 9% in patients without SVCS and 3% in those with the syndrome. When the malignancy is treated with radiation therapy, 46% of patients who have non-small cell lung cancer experience relief of symptoms, compared with 62% of patients who have small cell bronchogenic carcinoma. The 2-year survival rate of 5% is almost the same for both groups.[2]

Most patients with non-Hodgkin lymphoma and SVCS respond to appropriate chemotherapy or to combined-modality regimens.

Etiology and Physiology

Since William Hunter first described SVCS in 1757, the spectrum of underlying conditions associated with it has shifted from tuberculosis and syphilitic aneurysms of the ascending aorta to malignant disorders. Almost 95% of SVCS cases described in published modern series result from cancer. The most common cause is small cell bronchogenic carcinoma, followed by squamous cell carcinoma of the lung, adenocarcinoma of the lung, non-Hodgkin lymphoma, and large cell carcinoma of the lung.[3] Nonmalignant causes of SVCS in cancer patients include thrombosis that is associated with intracaval catheters or pacemaker wires.[4] A rare cause of SVCS is fibrosing mediastinitis, either idiopathic or associated with histoplasmosis.[5] Additional rare causes of SVCS include metastatic germ cell neoplasms, metastatic breast cancer, colon cancer, Kaposi sarcoma, esophageal carcinoma, fibrous mesothelioma, Behçet syndrome, thymoma, substernal thyroid goiter, Hodgkin lymphoma, and sarcoidosis.[6]

Knowledge of the anatomy of the SVC and its relationship to the surrounding lymph nodes is essential to understand development of the syndrome. The SVC is formed by the junction of the left and right brachiocephalic veins in the mid third of the mediastinum. The SVC extends caudally for 6 to 8 cm, coursing anterior to the right mainstem bronchus and terminating in the superior right atrium, and extends anteriorly to the right mainstem bronchus. The SVC is joined posteriorly by the azygos vein as it loops over the right mainstem bronchus and lies posterior to and to the right of the ascending aorta. The mediastinal parietal pleura is lateral to the SVC, creating a confined space, and the SVC is adjacent to the right paratracheal, azygous, right hilar, and subcarinal lymph node groups. The vessel itself is thin-walled, and the blood flowing in it is under low pressure. Thus, when the nodes or ascending aorta enlarge, the SVC is compressed, blood flow slows, and complete occlusion may occur.

The severity of SVCS depends on the speed of onset and location of the obstruction. The more rapid the onset, the more severe the symptoms because the collateral veins do not have time to distend to accommodate an increased blood flow.[7]

- If the obstruction is **above** the entry of the azygos vein, the syndrome is less pronounced because the azygous venous system can readily distend to accommodate the shunted blood with less venous pressure developing in the head, arms, and upper thorax.
- If the obstruction is **below** the entry of the azygos vein, more florid symptoms and signs are seen because the blood must be returned to the heart via the upper abdominal veins and the inferior vena cava, which requires higher venous pressure.

One study suggested that the general recruitment of venous collaterals over time may lead to remission of the syndrome, although the SVC remains obstructed.[8]

Assessment and Diagnosis

Once SVCS is recognized, prompt clinical attention is important. A diagnosis should be established before therapy begins because:[3]

- 75% of patients have symptoms and signs for longer than 1 week before seeking medical attention.

- Cancer patients diagnosed with SVCS do not die of the syndrome itself but rather their underlying disease.
- 3% to 5% of the patients diagnosed with SVCS do not have cancer.

In the absence of tracheal obstruction, SVCS is unlikely to be a life-threatening oncologic emergency, and treatment before definitive diagnosis is not justified.

The initial evaluation of the patient should include a chest x-ray to look for mediastinal masses and associated findings, such as pleural effusion, lobar collapse, or cardiomegaly. Computed tomography (CT) scanning of the thorax yields the most useful diagnostic information and can define the anatomy of the involved mediastinal nodes. Venous patency and the presence of thrombi are assessed using contrast and rapid scanning techniques.[9] Depending on local expertise, contrast or nuclear venography, magnetic resonance imaging, and ultrasonography may be valuable in assessing the site and nature of the obstruction.

If bronchogenic carcinoma is suspected, a sputum specimen should be obtained. If the specimen is negative, a biopsy specimen should be taken from the most accessible site that is clinically involved with disease. The biopsy approach depends on the working diagnosis, location of the tumor, physiological status of the patient, and expertise at the facility. It may include:[10]

- Bronchoscopy.
- Biopsy of palpable cervical or supraclavicular lymph nodes.
- Needle biopsy of a lung mass or mediastinal nodes using either CT or ultrasonography guidance.
- Mediastinoscopy.
- Mediastinotomy.
- Median sternotomy.
- Video-assisted thoracoscopy.
- Conventional thoracotomy.

The biopsy findings will help the clinician plan appropriate treatment.

Treatment Options for SVCS

The treatment of SVCS depends on the following:

- The etiology of the obstruction.
- The severity of symptoms.
- The patient's prognosis.
- Patient preferences and goals for therapy.

Radiation therapy or chemotherapy should be withheld until the etiology of the obstruction is clear. The treatments discussed here focus on SVC obstruction caused by a malignant tumor. Because the treatment of malignant obstruction may depend on tumor histology, a histological diagnosis should be made before starting treatment. Unless there is airway obstruction or cerebral edema, treatment may be delayed until after an assessment, with no negative effect on outcome.[1,11-15]

Medical management

A patient with sufficient collateral blood flow and minimal symptoms may not need treatment. If the lesion is above the azygous vein or if the onset of SVC occlusion is slow enough to allow sufficient collateral circulation, the symptoms and signs may stabilize, and further therapy may not be needed. Short-term palliation of a symptomatic patient who does not want aggressive treatment may be achieved by elevating the head and using corticosteroids and diuresis. Although

no definitive studies prove the effectiveness of steroids, they may be useful to treat respiratory compromise. Diuretics may relieve edema but can cause systemic complications, such as dehydration.[9,16]

Radiation therapy

If the obstruction of the SVC is caused by a tumor that is not sensitive to chemotherapy, radiation therapy should be given. Treatment with larger fractions of radiation is thought to help. However, one study shows no obvious need for large radiation fraction sizes for the first few radiation treatments as was previously believed.[17] Many fractionation schemes have been used, with doses ranging from 30 Gy in 10 fractions to 50 Gy in 25 fractions. Relief of symptoms is reported to be 62% to 80% in patients with small cell lung cancer and about 46% in patients with non-small cell lung cancer.[2,18] In one study, more than 90% of the patients achieved a partial or complete response with a 3-week regimen of 8 Gy given once a week for a total dose of 24 Gy.[19]

Chemotherapy

Chemotherapy is the treatment of choice for sensitive tumors such as lymphoma or small cell lung cancer. SVCS does not appear to be an independent prognostic factor, and its presence should not change the treatment approach. Rapid initiation of chemotherapy can result in complete and partial response rates of the SVCS of more than 80% in patients with small cell lung cancer.[2,18]

Thrombolysis

It has been suggested that SVCS arises when a thrombus forms in a partially occluded vein. In patients with a documented thrombus in the SVC, treatment may include thrombectomy, with or without tissue plasminogen activator or other thrombolytic agents, such as streptokinase or urokinase.[1]

Stent placement

Numerous small studies have reported the use of an intravascular expandable stent to reopen the occluded SVC; however, no prospectively designed comparative studies have been published.[20] The reported response rates have been about 90% or greater.[21][[Level of evidence: II](#)] There is no agreement on the need for anticoagulant therapy after stent placement. In one series that used anticoagulant therapy as part of the treatment protocol, there were reports of reocclusion after this therapy was stopped.[22][[Level of evidence: II](#)] However, in another study, 17 patients with cancer who were treated with stents and did not receive anticoagulant therapy had no occlusions.[23][[Level of evidence: II](#)]

Surgery

Surgical bypass of an obstructed SVC is more appropriate for patients with a benign obstruction than with a malignant obstruction,[24] although surgical bypass has also been used for patients with malignant obstructions.

Psychosocial Considerations

Patients and family members are often frightened and anxious about the symptoms of SVCS, particularly swelling, dysphagia, coughing, and hoarseness. Patients and family members need information about the cause of the symptoms and short-term measures for palliation, especially during the diagnostic period. When aggressive treatment is declined because of the terminal nature of the underlying disease, symptom management can be offered to patients and family members.

Because most adult patients who develop SVCS have lung cancer, the treatment and psychological support approaches for SVCS should take into account the patient's prognosis and mental condition, goals of care, and other symptoms caused by the malignancy.[25]

Pediatric Considerations

As described in this summary, SVCS refers to the symptoms associated with the compression or obstruction of the SVC; the compression of the trachea is termed superior mediastinal syndrome (SMS). Because SMS and the resulting

respiratory compromise frequently occur in children with SVCS, the two syndromes have become almost synonymous in pediatric practice.[26,27] In adults, the trachea and the right mainstem bronchus are relatively rigid structures compared with the vena cava, but in children, these structures are more susceptible to compression. In addition, the relatively smaller intraluminal diameters of a child's trachea and bronchus can tolerate little edema before respiratory symptoms occur. Because of this accompanying respiratory component, SVCS in children differs from the adult syndrome and is a serious medical emergency.

The most common symptoms of SVCS in children are similar to those in adults and include the following:[26]

- Coughing.
- Hoarseness.
- Dyspnea.
- Orthopnea.
- Chest pain.

Less common but more serious symptoms include the following:

- Syncope.
- Anxiety.
- Confusion.
- Lethargy.
- Headache.
- Distorted vision.
- A sense of fullness in the ears.

SVCS is rare in children, appearing at presentation in 12% of pediatric patients with malignant mediastinal tumors.[28,29] The etiology, diagnosis, and treatment of SVCS in children differ from those in adults. The most frequent cause of SVCS in adults is bronchogenic carcinoma;[3] in children, it is non-Hodgkin lymphoma. As in adults, a frequent nonmalignant cause is thrombosis from catheterization for venous access.[26]

A physical examination, chest radiograph, and medical history are usually sufficient to diagnose SVCS. If lymphomas or other malignant diseases are suspected, a tissue sample may aid diagnosis. However, the procedure to obtain the specimen may involve significant risk and may not be clinically feasible. Children with SVCS have a poor tolerance for the necessary general anesthesia because the accompanying cardiovascular and pulmonary changes aggravate the SVCS, often making intubation impossible. Also, extubation may be difficult or impossible, requiring prolonged airway provision (intubation). A CT scan of the chest to determine tracheal size, upright and supine echocardiography, and a flow volume loop may help evaluate anesthetic risk. Because anesthesia use is a serious risk, the diagnosis should be made with the least invasive means possible.[30] Published reports suggest a stepwise approach.[26]

When a malignant mass causes SVCS, the situation may be a medical emergency with no opportunity to establish a tissue diagnosis. In these cases, the most appropriate course may be to initiate empiric therapy before biopsy. The traditional empiric therapy is radiation therapy, with the daily dose governed by the presumed radiosensitivity of the tumor. After radiation therapy, respiratory deterioration from the apparent tracheal swelling may occur. This swelling occurs because narrow lumens in children cannot accommodate edema and because of the greater degree of edema at onset, the result of rapid tumor growth in children. In these situations, a course of prednisone at 10 mg/m² of body surface area 4 times per day may be necessary.[26]

In addition to radiation, empiric therapy for SVCS has included chemotherapeutic agents incorporating steroids, cyclophosphamide, or both in combination with an anthracycline and vincristine.[26] If the tumor fails to respond, it

may be a benign lesion.

If surgery becomes necessary, it should be performed with the patient in the semi-Fowler's position, allowing the surgeon the ability to rapidly change the patient's position to lateral or prone. Cardiopulmonary bypass facilities and a rigid bronchoscope should be available in a standby capacity.[30]

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The PDQ cancer information summaries are reviewed regularly and updated as new information becomes available. This section describes the latest changes made to this summary as of the date above.

Editorial changes were made to this summary.

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About This PDQ Summary

Purpose of This Summary

This PDQ cancer information summary for health professionals provides comprehensive, peer-reviewed, evidence-based information about the pathophysiology and treatment of cardiopulmonary syndromes, including dyspnea, malignant pleural effusion, malignant pericardial effusion, and superior vena cava syndrome. It is intended as a resource to inform and assist clinicians in the care of their patients. It does not provide formal guidelines or recommendations for making health care decisions.

Reviewers and Updates

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Board members review recently published articles each month to determine whether an article should:

- be discussed at a meeting,
- be cited with text, or
- replace or update an existing article that is already cited.

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Updated: February 20, 2025

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